

RIVNE INTL, Ukraine

WMO: 333010

Lat: 50.6025N

Lon: 26.1528E

Elev: 231

StdP: 98.58

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-15.3	-21.4	0.6	-18.7	-17.9	0.8	-15.1	12.9	2.4	11.9	1.8	2.6	110	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.7	20.0	28.9	19.3	27.1	18.7	21.5	27.7	20.5	26.4	19.7	25.0	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.6	24.3	18.6	13.8	23.0	17.8	13.1	22.1	63.7	27.9	60.1	26.3	57.1	25.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.2	8.1	DB	-22.4	33.0	4.1	1.4	-25.3	34.0	-27.7	34.8	-30.0	35.6	-32.9	36.7	(4)
(5)			WB	-22.5	23.2	3.9	0.9	-25.4	23.9	-27.7	24.4	-29.9	24.9	-32.7	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-3.8	-2.4	2.0	9.2	14.6	17.7	19.6	18.9	13.8	8.2	3.0	-1.9	(6)
(7)		DBStd	9.48	5.89	5.82	4.83	4.37	4.09	3.51	3.26	3.34	3.87	4.38	4.78	5.27	(7)
(8)		HDD10.0	1777	429	348	248	66	8	0	0	0	10	87	213	368	(8)
(9)		HDD18.3	3836	688	581	506	276	129	54	24	34	145	313	459	626	(9)
(10)		CDD10.0	1156	0	0	1	41	150	231	299	275	123	32	4	0	(10)
(11)		CDD18.3	172	0	0	0	1	12	36	65	51	8	0	0	0	(11)
(12)		CDH23.3	1637	0	0	0	13	139	302	583	505	94	2	0	0	(12)
(13)		CDH26.7	457	0	0	0	2	25	65	181	165	19	0	0	0	(13)
(14)	Wind	WSAvg	4.0	4.7	4.4	4.7	4.2	3.8	3.6	3.4	3.2	3.6	3.9	4.4	4.6	(14)
(15)	Precipitation	PrecAvg	612	32	31	33	40	64	74	97	59	59	44	38	40	(15)
(16)		PrecMax	816	72	60	75	117	146	132	189	127	164	115	77	75	(16)
(17)		PrecMin	404	12	15	2	6	25	33	24	4	5	3	4	15	(17)
(18)		PrecStd	103	14	11	17	23	27	23	53	34	37	28	16	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	10.9	16.8	25.1	29.3	30.5	32.8	33.1	28.9	23.1	16.1	9.2	(19)
(20)			MCWB	6.0	7.7	10.0	13.9	17.7	20.3	21.1	19.8	17.5	15.3	11.9	7.6	(20)
(21)		2%	DB	6.0	7.5	13.4	21.6	26.9	28.6	30.8	30.6	26.0	19.9	13.2	7.2	(21)
(22)			MCWB	4.6	5.6	8.1	12.7	16.9	19.4	20.7	19.5	16.5	14.2	10.6	6.2	(22)
(23)		5%	DB	4.2	5.8	11.0	19.1	24.8	26.9	28.8	28.4	23.4	17.0	11.2	5.5	(23)
(24)			MCWB	3.1	4.3	6.9	11.5	16.3	18.8	20.0	19.1	16.0	12.9	9.6	4.6	(24)
(25)		10%	DB	2.8	4.0	8.7	16.9	22.3	24.9	26.8	26.1	20.9	14.8	9.4	4.0	(25)
(26)			MCWB	2.0	2.8	5.8	10.6	15.1	18.0	19.3	18.4	15.2	11.7	8.1	3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	8.0	10.5	15.4	19.8	21.8	23.2	22.7	19.4	16.7	12.4	8.1	(27)
(28)			MCDWB	7.5	10.4	15.3	23.2	26.1	27.3	29.4	29.5	26.8	21.1	14.9	9.1	(28)
(29)		2%	WB	4.7	5.8	8.9	13.9	18.2	20.5	21.8	21.3	17.9	14.9	11.1	6.3	(29)
(30)			MCDWB	5.6	7.2	12.7	19.4	24.2	26.4	28.0	27.4	22.8	18.8	12.8	7.0	(30)
(31)		5%	WB	3.2	4.4	7.4	12.5	17.0	19.6	20.8	20.1	16.7	13.5	9.7	4.7	(31)
(32)			MCDWB	4.1	5.5	10.0	17.5	23.1	25.1	26.8	25.4	21.4	16.5	11.2	5.3	(32)
(33)		10%	WB	2.0	2.8	6.0	11.3	15.9	18.7	19.9	19.0	15.6	12.0	8.1	3.3	(33)
(34)			MCDWB	2.6	3.8	8.2	15.7	20.8	23.5	25.1	24.4	20.0	14.4	9.2	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.6	5.6	7.3	10.0	10.7	10.4	10.6	11.2	9.9	8.2	5.0	4.3	(35)
(36)			MCDBR	4.6	6.8	11.5	13.7	13.9	12.9	13.5	14.5	14.3	12.0	8.3	5.4	(36)
(37)			MCWBR	4.0	5.2	7.3	6.4	6.0	5.6	5.3	5.6	6.3	6.9	5.8	4.6	(37)
(38)		5% WB	MCDWR	4.3	6.5	9.9	12.1	12.1	11.3	11.9	12.3	12.0	10.6	7.6	5.2	(38)
(39)			MCWBR	3.8	5.2	6.8	6.2	6.0	5.6	5.3	5.6	6.1	6.5	5.5	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.341	0.378	0.428	0.412	0.406	0.436	0.434	0.400	0.362	0.343	0.307	(40)	
(41)		taud	2.221	2.172	2.188	2.134	2.250	2.289	2.219	2.221	2.270	2.358	2.310	2.274	(41)	
(42)		Ebn at Noon	695	769	811	809	845	852	817	797	782	747	655	643	(42)	
(43)		Edh at Noon	73	99	118	140	130	126	133	127	108	83	68	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.85	1.67	2.84	4.24	5.35	5.73	5.44	4.86	3.32	1.97	0.85	0.59	(44)	
(45)		RadStd	0.15	0.26	0.40	0.43	0.49	0.42	0.44	0.32	0.49	0.32	0.09	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.53	N/A	N/A	+1.26	N/A	N/A	N/A	-174	+124	+52
(47) Regional (8 neighbors)		+0.66	N/A	N/A	+0.90	+0.54	+0.44	N/A	-200	+114	+45

Nomenclature: See separate page